Machine Learning
Multiple Linear Regression

From Prediction to Learning

Weights, Bias, Loss Function, Cost Function and Gradient Descent

Key Idea

A machine learning model learns suitable parameter values from training data. The basic learning process is:

$$\boxed{\text{Prediction} \rightarrow \text{Error} \rightarrow \text{Cost} \rightarrow \text{Parameter Update} \rightarrow \text{Improved Prediction}}$$

1 From Simple to Multiple Linear Regression

In Simple Linear Regression, one input feature is used to predict a continuous target.

$$\hat{y} = w_0 + w_1 x \tag{1}$$

where:

- x = input feature
- $\hat{y}$ = predicted value
- w_1 = weight
- w_0 = bias or intercept

When multiple features are available, we use Multiple Linear Regression.

$$\boxed{\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p} \tag{2}$$

where:

- $x_1, x_2, \ldots, x_p$ are input features
- $w_1, w_2, \ldots, w_p$ are corresponding weights
- w_0 is the bias
- $\hat{y}$ is the predicted output

Key Idea

Multiple Linear Regression uses multiple input features to predict one continuous target variable.

2 Example: House Price Prediction

Suppose we want to predict the price of a house.

Possible features are:

Feature	Example Value
Area	1000 sq.ft
Number of Bedrooms	2
Age of House	5 years
Distance from Metro	2 km

The model can be written as:

$$\hat{y} = w_0 + w_1(\text{Area}) + w_2(\text{Bedrooms}) + w_3(\text{Age}) + w_4(\text{Distance}) \tag{3}$$

The question is:

Where do the values of the weights and bias come from?

Key Idea

The model must **learn** the values of its parameters from the training data.

3 Weights

Consider the model:

$$\hat{y} = 10 + 0.05(\text{Area}) + 5(\text{Bedrooms}) - 2(\text{Age}) \tag{4}$$

The parameters are:

Parameter	Value
Bias w_0	10
Area weight w_1	0.05
Bedroom weight w_2	5
Age weight w_3	-2

A weight determines how a feature contributes to the model's prediction.

- Positive weight $\rightarrow$ prediction tends to increase.
- Negative weight $\rightarrow$ prediction tends to decrease.

3.1 Understanding a Weight

Suppose the area increases by 100 sq.ft.

The weight of Area is:

$$w_1 = 0.05$$

Therefore, the change in prediction due to Area is:

$$\Delta\hat{y} = 0.05 \times 100 = 5 \tag{5}$$

Therefore:

Key Idea

If Area increases by 100 units, the model's prediction increases by 5 units, assuming the other features remain unchanged.

4 Bias

For the model:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 \tag{6}$$

w_0 is called the **bias** or **intercept**.

It represents the baseline value of the prediction before considering the contribution of the input features.

$$\boxed{\text{Prediction} = \text{Bias} + \text{Feature Contributions}} \tag{7}$$

5 The Model Has to Learn

Initially, the model does not know the best values of:

$$w_0, w_1, w_2, \ldots, w_p$$

The model may start with initial parameter values.

It then:

$$\boxed{\text{Parameters} \rightarrow \text{Prediction} \rightarrow \text{Error}}$$

The next question is:

How do we know whether the prediction is good or bad?

6 Prediction Error

For one training observation:

$$y = \text{Actual Value}$$

$$\hat{y} = \text{Predicted Value}$$

The prediction error is:

$$\boxed{e = y - \hat{y}} \tag{8}$$

For example:

$$y = 120$$

$$\hat{y} = 105$$

Therefore:

$$e = 120 - 105 = 15 \tag{9}$$

The prediction differs from the actual value by 15 units.

7 Loss Function

A loss function measures how wrong the prediction is for an individual training example.
For squared error:

$$\boxed{L(y, \hat{y}) = (y - \hat{y})^2} \tag{10}$$

For:

$$y = 120, \qquad \hat{y} = 105$$

we get:

$$L = (120 - 105)^2 \tag{11}$$
$$= 15^2 \tag{12}$$
$$= 225 \tag{13}$$

Therefore, the loss for this observation is:

$$\boxed{225}$$

Key Idea

$$\boxed{\text{Loss} = \text{Error for one training example}}$$

8 Why Square the Error?

Squared error is commonly used because:

1. Positive and negative errors do not cancel each other.
2. Larger errors receive a larger penalty.

For example:

$$(-2)^2 = 4$$

whereas:

$$(-10)^2 = 100$$

9 From Loss to Cost Function

A training dataset contains many observations.
Each observation has its own loss:

$$L_1, L_2, L_3, \ldots, L_n$$

We need a single value representing the overall error of the model.
For Mean Squared Error:

$$\boxed{J = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{14}$$

where:

- n = number of training observations
- y_i = actual value for observation i
- $\hat{y}_i$ = predicted value for observation i

Key Idea

Loss → One Observation

Cost → Overall Training Dataset

10 Connection with MSE

You have already studied Mean Squared Error (MSE) as an evaluation metric.

The mathematical expression is:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{15}$$

The same quantity can be used as a cost function during training.

The important difference is the purpose.

During Training	After Training
Used as a cost to guide parameter updates	Used as an evaluation metric

Remember

The mathematical expression can be the same, but its role can be different depending on the context.

11 Goal of Training

Different parameter values produce different predictions.

Different predictions produce different costs.

Therefore, the objective of training is:

$$\boxed{\min J} \tag{16}$$

In words:

Find parameter values that minimize the cost.

12 Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the cost function.

Imagine the cost function as a landscape.

The goal is to reach the lowest point.

Goal: Move toward minimum cost

The basic process is:

$$\boxed{\text{Start} \rightarrow \text{Check Direction} \rightarrow \text{Take a Step} \rightarrow \text{Repeat}}$$

13 What is a Gradient?

For a parameter w_j, the gradient is represented by:

$$\frac{\partial J}{\partial w_j} \tag{17}$$

It tells us how the cost changes when the parameter w_j changes.
In simple terms:

Key Idea

The gradient tells us the direction in which the cost increases.
Therefore, Gradient Descent moves in the opposite direction to reduce the cost.

14 Gradient Descent Update Rule

The general update rule for a parameter is:

$$\boxed{w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}} \tag{18}$$

where:

- w_j = current value of the parameter
- α = learning rate
- J = cost function
- $\frac{\partial J}{\partial w_j}$ = gradient of the cost with respect to w_j

Key Idea

The model updates its parameters repeatedly so that the cost becomes smaller.

15 Learning Rate

The learning rate is represented by:

$$\alpha$$

It controls the size of each update.

Learning Rate Too Small

Very small steps

The learning process may be very slow.

Learning Rate Too Large

Very large steps

The model may overshoot the minimum.

Suitable Learning Rate

The model gradually approaches a minimum.

Remember

$\boxed{\alpha = \text{Step Size}}$

16 Complete Learning Process

The complete learning cycle is:

Training Data
↓
Initialize Weights and Bias
↓
Make Predictions
↓
Calculate Loss
↓
Calculate Cost
↓
Calculate Gradient
↓
Update Parameters
↓
Repeat
↓
Trained Model

17 Training vs Evaluation

It is important to distinguish training from evaluation.

During Training

$\boxed{\text{Prediction} \rightarrow \text{Cost} \rightarrow \text{Gradient Descent} \rightarrow \text{Parameter Update}}$

This process is repeated.

After Training

The trained model is evaluated using metrics such as:

$$MAE, \qquad MSE, \qquad R^2$$

Key Idea

Training asks:
How can I improve the model?

Evaluation asks:
How well does the trained model perform?

18 Final Summary

Features → Weights + Bias
→ Prediction
→ Loss
→ Cost
→ Gradient Descent
→ Updated Parameters
→ Repeat

The model learns by minimizing the cost function.

19 Quick Revision

Term	Meaning
Feature	Input variable used by the model
Weight	Controls the contribution of a feature
Bias	Baseline/intercept of the model
Prediction	Output produced by the current model
Loss	Error for one observation
Cost	Overall error across the training data
Gradient	Shows how cost changes with respect to a parameter
Learning Rate	Controls the size of parameter updates
Gradient Descent	Optimization algorithm used to minimize cost